Postgre 192.168.56.10
Backend 192.168.56.11
Redis 192.168.56.12
Proxy 192.168.56.13

Успешный запрос:

```
[proxy@localhost ~]$ curl http://192.168.56.13:5000/user?id=5
{"cached":false,"user":{"age":35,"id":5,"name":"Charlie Davis"}}
[proxy@localhost ~]$ curl http://192.168.56.13:5000/user?id=5
{"cached":true,"user":{"age":35,"id":5,"name":"Charlie Davis"}}
```

Проверка iptables (запросы не идут):

```
[redis@localhost ~]$ curl http://192.168.56.11:8080/user?id=5
^C
```

```
[proxy@localhost ~]$ curl http://192.168.56.10
^C
```

```
postgre@vm-postgres:~$ curl http://192.168.56.12
^Cpostgre@vm-postgres:~$
```

Проверка служб:

```
gofer@vm-golang:~$ sudo systemctl status backend-api
● backend-api.service - Backend API
     Loaded: loaded (/usr/lib/systemd/system/backend-api.service; enabled; pres>
     Active: active (running) since Tue 2026-04-14 19:12:38 MSK; 10min ago
   Main PID: 1342 (backend-api)
      Tasks: 5 (limit: 4483)
     Memory: 1.3M (peak: 1.7M)
        CPU: 7ms
     CGroup: /system.slice/backend-api.service
             └─1342 /usr/bin/backend-api

апр 14 19:12:33 vm-golang systemd[1]: backend-api.service: Failed with result '>
апр 14 19:12:38 vm-golang systemd[1]: backend-api.service: Scheduled restart jo>
апр 14 19:12:38 vm-golang systemd[1]: Started backend-api.service - Backend API.
апр 14 19:12:38 vm-golang backend-api[1342]: 2026/04/14 19:12:38 Backend API за>
```

```
● cache-api.service - Cache API Proxy
     Loaded: loaded (/usr/lib/systemd/system/cache-api.service; enabled; preset: disabled)
     Active: active (running) since Tue 2026-04-14 15:13:32 CEST; 3h 10min ago
   Main PID: 834 (python3)
      Tasks: 1 (limit: 22355)
     Memory: 44.0M (peak: 44.1M)
        CPU: 452ms
     CGroup: /system.slice/cache-api.service
             └─834 /usr/bin/python3 /usr/bin/cache-api.py

апр 14 15:13:32 localhost.localdomain systemd[1]: Started Cache API Proxy.
апр 14 15:13:35 localhost.localdomain python3[834]:  * Serving Flask app 'cache-api'
апр 14 15:13:35 localhost.localdomain python3[834]:  * Debug mode: off
апр 14 15:13:35 localhost.localdomain python3[834]: WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI se>
апр 14 15:13:35 localhost.localdomain python3[834]:  * Running on all addresses (0.0.0.0)
апр 14 15:13:35 localhost.localdomain python3[834]:  * Running on http://127.0.0.1:5000
апр 14 15:13:35 localhost.localdomain python3[834]:  * Running on http://10.0.2.15:5000
апр 14 15:13:35 localhost.localdomain python3[834]: Press CTRL+C to quit
апр 14 18:19:54 localhost.localdomain python3[834]: 192.168.56.13 - - [14/Apr/2026 18:19:54] "GET /user?id=5 HTTP/1.1" 200 -
апр 14 18:19:58 localhost.localdomain python3[834]: 192.168.56.13 - - [14/Apr/2026 18:19:58] "GET /user?id=5 HTTP/1.1" 200 -
```

```
postgre@vm-postgres:~$ sudo systemctl status postgresql
● postgresql.service - PostgreSQL RDBMS
     Loaded: loaded (/usr/lib/systemd/system/postgresql.service; enabled; prese>
     Active: active (exited) since Tue 2026-04-14 16:13:32 MSK; 9min ago
    Process: 1365 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
   Main PID: 1365 (code=exited, status=0/SUCCESS)
        CPU: 3ms

апр 14 16:13:32 vm-postgres systemd[1]: Starting postgresql.service - PostgreSQ>
апр 14 16:13:32 vm-postgres systemd[1]: Finished postgresql.service - PostgreSQ>
```

```
● redis.service - Redis persistent key-value database
     Loaded: loaded (/usr/lib/systemd/system/redis.service; enabled; preset: disabled)
    Drop-In: /etc/systemd/system/redis.service.d
             └─limit.conf
     Active: active (running) since Tue 2026-04-14 16:13:31 MSK; 3h 15min ago
   Main PID: 943 (redis-server)
     Status: "Ready to accept connections"
      Tasks: 6 (limit: 22355)
     Memory: 11.4M (peak: 12.4M)
        CPU: 750ms
     CGroup: /system.slice/redis.service
             └─943 "/usr/bin/redis-server 127.0.0.1:6379"

апр 14 16:13:31 localhost.localdomain systemd[1]: Starting Redis persistent key-value database...
апр 14 16:13:31 localhost.localdomain systemd[1]: Started Redis persistent key-value database.
```

Репа:

https://github.com/KonservnayaBaka/finalBaseProject